

1. The list implementation I used as the snake's body is the ArrayList
2. I selected that implementation because I personally currently prefer the Array list to the Linked List since I am more familiar with it and know more about how to make the simple functions than the Linked List functions.
3. The list operations that occur during a normal snake movement are the addElement and removeElement, specifically we are wanting to change the first position based on the movement inserted and remove the last trace of the snake when moving.
4. The operations that occur when the snake eats the food is addElement and we have the position we add set to 0, the beginning, since we want to add more of the snake.
5. The worst-case runtime for one movement step is that we might leave traces of the snake behind and not update.
6. Collision detection affects the runtime of a game step by alerting the user that the game is over and will no longer display the board and ask for user input on where to go next.
7. The advantages that the array-based implementation provides are that it requires less memory compared to a Node since they are dynamically allocated and connected using pointers.
8. The advantages that the link list provides are that they have non-contiguous memory locations for each of the nodes involved, they can grow and shrink at runtime which means that we don't need to give them an initial size, and growing and shrinking during runtime also means that there will be no memory waste.
9. The implementation I would choose if indexed access was the most common operation would be the Array-based implementation since it only requires less memory to obtain the access.
10. The implementation I would choose if insertion and removal at the front were the most common operations would be the linked implementation because it only requires just two lines of code in order to insert or remove an element, while in array-based implementation, you would have to manually change all of the elements that have been affected by the insertion or removal.